# Jesús M. Rueda-Becerril | Ph.D.

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## Professional Summary

Computational Scientist (PhD, Physics) with 10+ years building high-performance numerical algorithms and physics-based simulations. Expertise in scientific machine learning and geospatial data analysis. Architect of open-source simulation libraries, leader of funded research projects, and translator of complex physical models into production-quality scientific software across cross-functional research and engineering teams.

## Skills

**Languages**: Python, C/C++, R, Rust, SQL, Fortran 95, Java, Julia, Shell

**ML & Data**: Scientific machine learning, PyTorch, scikit-learn, TensorFlow, random forest, neural networks, SVM, gradient-based optimization, ML-assisted parameter estimation, NumPy, SciPy, Pandas, Astropy

**Geospatial**: Sentinel-1/2 multispectral processing, raster/vector data pipelines, QGIS

**HPC & Parallel**: OpenMP, OpenACC (GPU offloading), MPI, SLURM, PBS, HDF5

**Tools**: Git, Docker, Azure, Jenkins, Kafka, Jira, ParaView, VisIt

## Experience

**Independent Contractor,** *Mercor* **Jan 2026 – Present**

Physics subject-matter expert evaluating frontier language models for a leading AI lab.

- Review 50+ expert-level problems in physics, hydrodynamics, astrophysics, and cosmology; write original prompts that stump frontier models.

**Spatial Data Scientist,** *TealWaters* **May 2025 – Jan 2026**

Built geospatial ML pipelines and optimized terrain-analysis code for wetland mapping.

- Optimized the terrain-modeling code that computes elevation derivatives from 1 m, 3 m, and 10 m DEMs over 20 tiles, improving its computational performance.
- Mapped wetland probability across the 2,160 km² Skykomish watershed with the Wetland Intrinsic Potential (WIP) random-forest model, reaching >75% accuracy.

**Independent Researcher & Open-Source Developer,** *Self-directed* **Jan 2024 – May 2025**

Continued simulation research and open-source development between industry roles.

- Co-developed Tleco, a Rust/Python toolkit that solves the Fokker–Planck equation coupled to radiative transfer for relativistic outflows (ApJ 2024).
- Developing WindsOfChange with the UW Forest Mycobiome Lab: a C++/R library for wind-driven spore dispersal over hilly forest terrain (RcppParallel, OpenMP).

**Software Engineer,** *Paychex* **Apr 2022 – Jan 2024**

Built and tested data-integration services between enterprise systems and Oracle E-Business Suite.

- Developed and tested production data-pipeline software (Java, Python, SQL) with CI/CD.

**Postdoctoral Research Associate,** *Rochester Institute of Technology* **Feb 2021 – Apr 2022**

HPC simulations of supermassive black hole binaries in an NSF-funded project.

- Led a 7-person team upgrading PatchworkMHD, a C code for MHD simulations of supermassive black hole binaries; added black hole spin with under 5% runtime overhead.
- Ran and benchmarked large production simulations on Frontera (TACC), about 10,000 node-hours, and identified performance bottlenecks in the parallel code.
- Calibrated Paramo against Fermi-LAT observations with gradient-based optimization that updates the model automatically; supervised a graduate student on the same problem.
- Collaborated on a multi-institutional HPC project on binary neutron star mergers, resulting in 2 publications.

**Postdoctoral Research Fellow,** *Purdue University* **Oct 2018 – Nov 2020**

Radiative-transfer modeling of relativistic jets (blazars and gamma-ray burst afterglows).

- Designed and wrote Paramo, an open-source Fortran 95 library that solves the Fokker–Planck equation for relativistic particle distributions coupled to radiative transfer; optimized it with OpenMP and tested GPU offload with OpenACC.
- Implemented implicit PDE solvers, numerical integration, and arbitrary-shape particle distributions without slowing the simulation.
- Wrote Python tools to analyze Paramo output and run parameter optimization; other groups have adopted Paramo, including Combi & Siegel (2023).
- Served as co-I on two NASA Fermi Guest Investigator proposals, one funded ($68K) to model blazar radiation; mentored 3 graduate students and published 2 peer-reviewed papers.

**Postdoctoral Research Fellow,** *Universidad Michoacana de San Nicolás de Hidalgo* **Jan 2018 – Sep 2018**

- Developed a Python pipeline to process black hole simulation outputs from the GRTrans numerical code, generating SVM training data for radio image prediction.
- Built open-source data analysis and visualization tools in Python for radiative transfer phenomena in relativistic astrophysics scenarios.

**Graduate Research Assistant,** *Universitat de València* **Oct 2011 – Jul 2017**

- Studied how magnetization of shocked plasma shapes the radiation spectra of relativistic internal shocks, using the C-SPEV simulation code.

## Projects

**Blazar Population ML** **Jan 2026 – Present**

*Scientific ML (Python, scikit-learn, PyTorch)* github.com/altjerue/ilwikak-ml

- Applying classical ML and Bayesian inference to the Fermi-LAT 4LAC-DR3 catalog (3,407 sources) to recover physical jet parameters from multi-wavelength data.
- Building a neural-network emulator of the physics-based Tleco model that maps physical jet parameters to observable spectral energy distributions, to be validated against Tleco output (in progress).
- Forecasting blazar gamma-ray flares with a time-series foundation model fine-tuned on Tleco-generated synthetic light curves.

## Selected Publications (9 peer-reviewed articles, 3 first-author)

- Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Tleco: A Toolkit for Modeling Radiative Signatures from Relativistic Outflows*, ApJ 976, 182 (2024), arXiv:2405.17581.
- **Rueda-Becerril, J. M.**, Harrison, A. O., & Giannios, D. *Blazar jets launched with similar energy per baryon, independently of their power*, MNRAS 501, 4092–4102 (2021), arXiv:2009.02273.
- **Rueda-Becerril, J. M.**, Mimica, P., & Aloy, M. A. *On the influence of a hybrid thermal–non-thermal distribution in the internal shocks model for blazars*, MNRAS 468, 1169–1182 (2017), arXiv:1612.06383.

## Education

**Universitat de València,** *Ph.D. in Physics, Excellent Cum Laude* **Jul 2017**

**Universidad Michoacana de San Nicolás de Hidalgo,** *M.Sc. in Physics* **Sep 2011**

**Universidad Autónoma del Estado de México,** *B.Sc. in Physics* **Dec 2008**